1. site selection & scanning

Selecting my site was difficult, as my studio project is siteless. The site chosen is in the Cannery District, and I chose this because it has interesting urban growth and the highway that runs by it, elevated by means of topographic infill.

This part of the chosen site was particularly interesting because it involves a **bridge** for interstate-90. The interstate is slowly **built up in elevation through berms**, which was another interesting layer of topography to an **otherwise flat site.**

berming + bridge

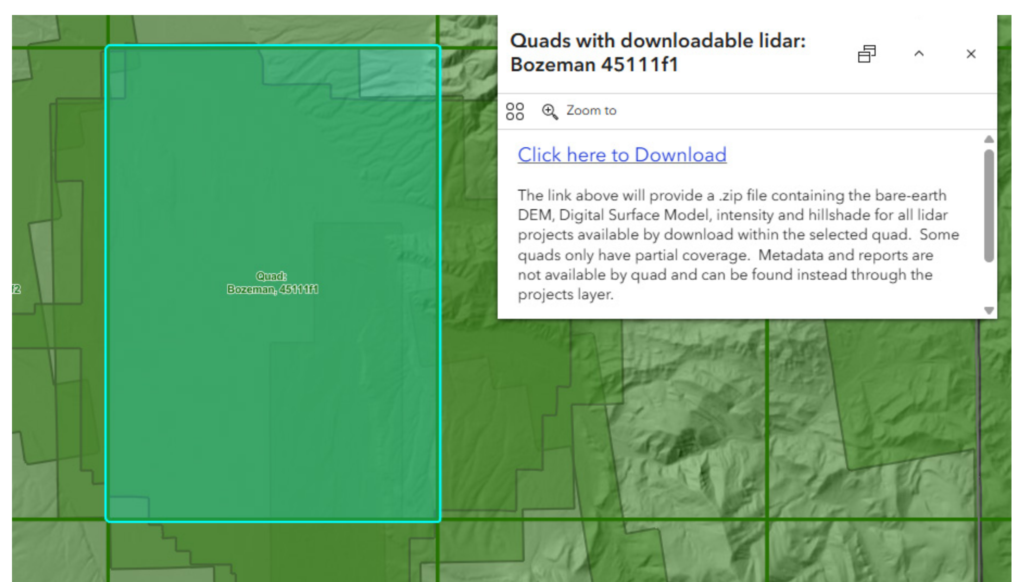
The **baseball field** on the chosen topographic area was another point of interest, as it needs to be **quite flat for a proper playing surface**. It was curious to see just how flat the field was according to LiDAR data, and **how it compared to the rest of the area**.

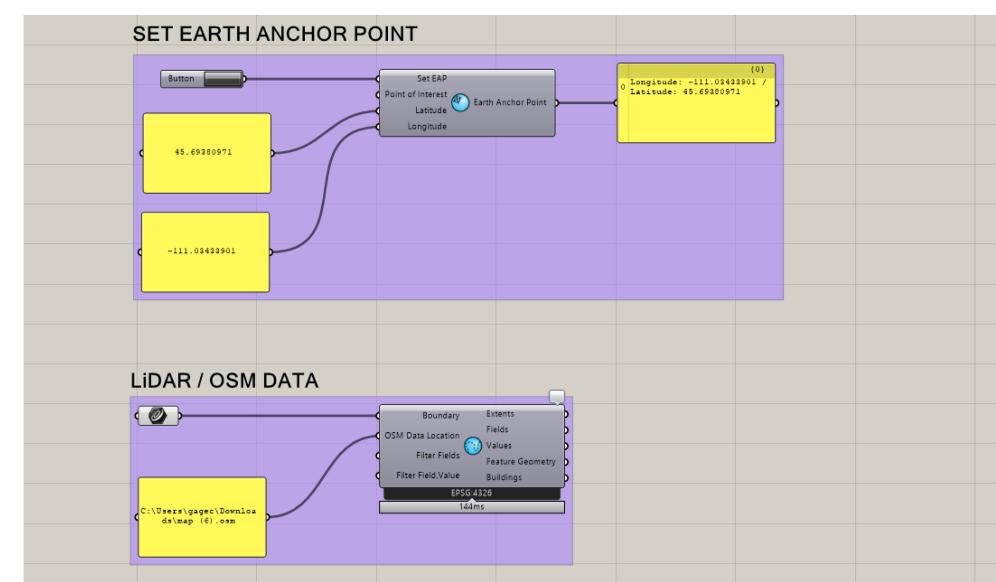
baseball field

data acquisition + grasshopper

Acquiring the data was successful, as Montana LiDAR Inventory has open source data for most areas. This provided accurate topographic data along with informaiton about parcels, roads, and other things specific to Bozeman.

The initial Grasshopper scripts were straightforward, with the first one establishing the Earth Anchor Point, so data can coordinate itself about that center. The second part was for importing OSM data which is is the foundation for the rest of the information.



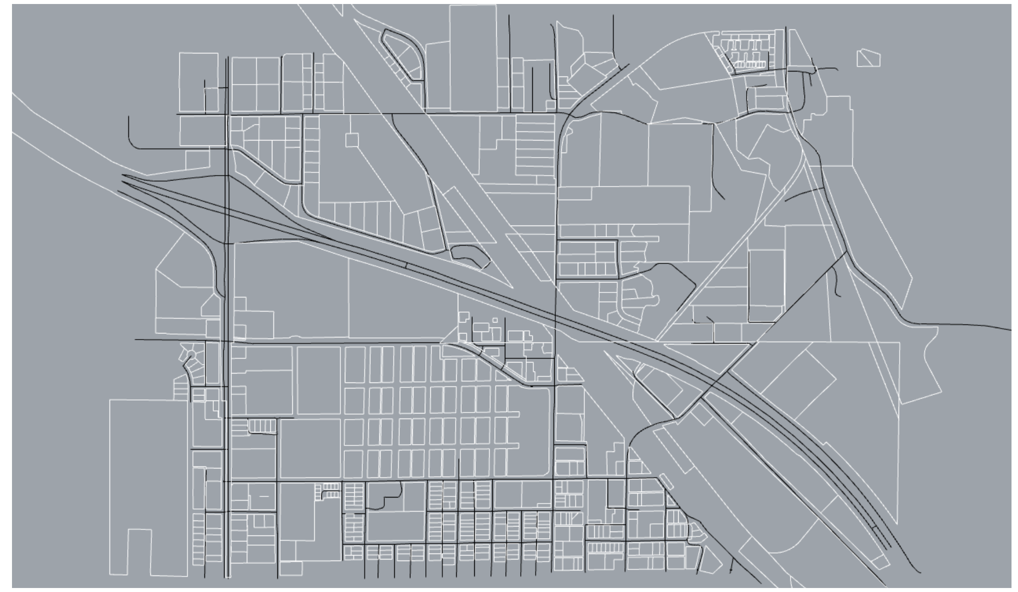


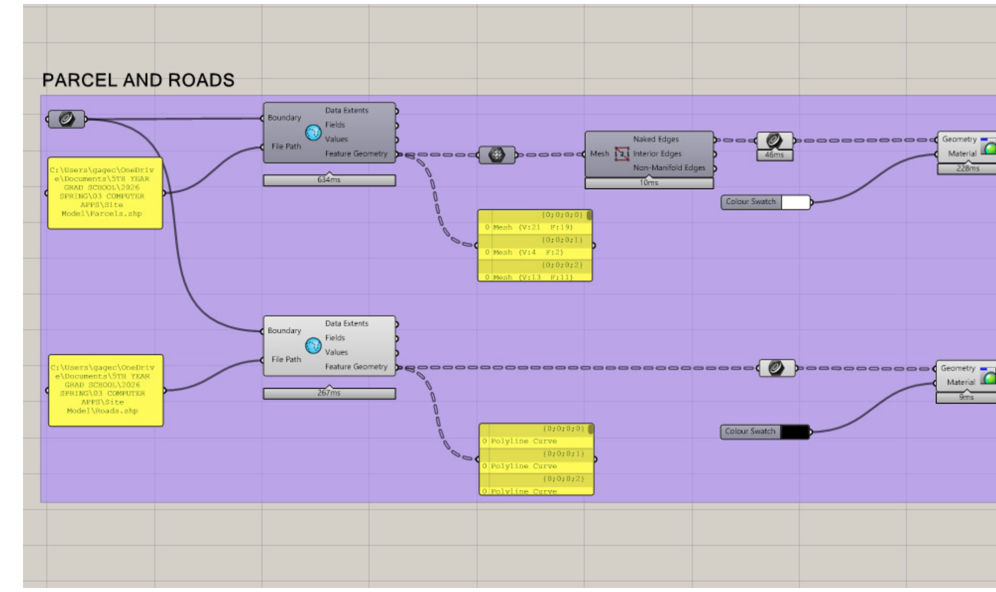
assignment 01 - Site Modeling with LiDAR and Physical Fabrication

2. data processing & digital modeling

parcels and roads

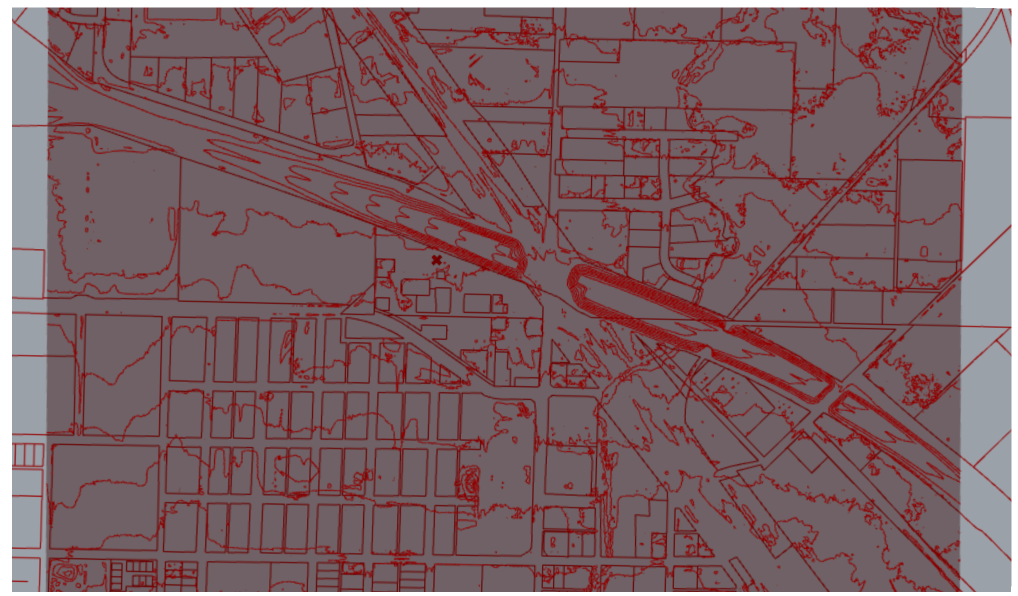
Processing this data from OSM data took many subscripts to digest the information and lay it out in a meangful manner. The next steps involved **extracting parcel information** to create an outline for an understanding of land dispersion. Adding roads as linework on top of this helped create an understanding of **key circulation and traffic patterns** around the site, which **could inform entrance layouts in future projects.**

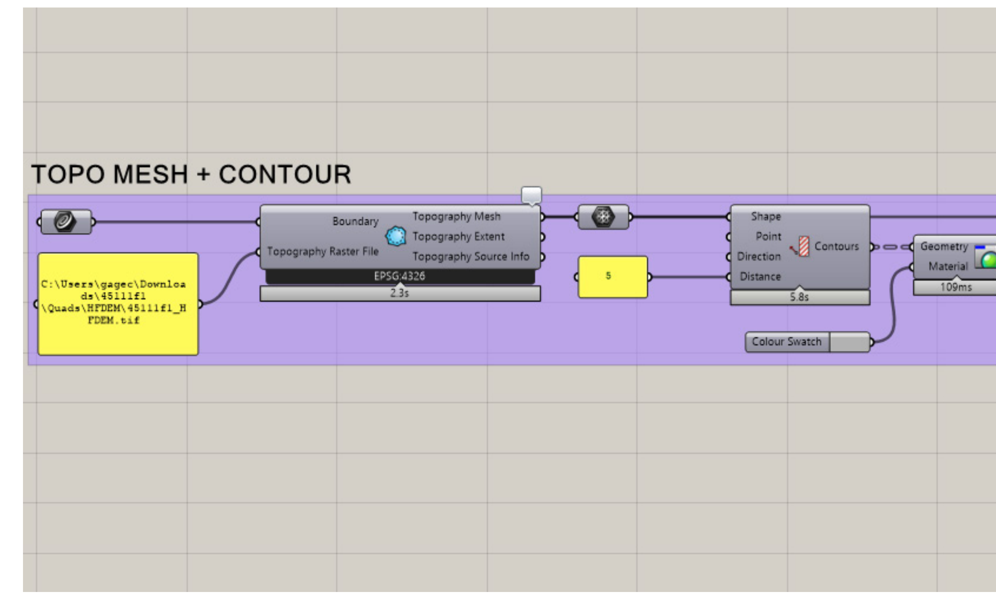




topo mesh + contours

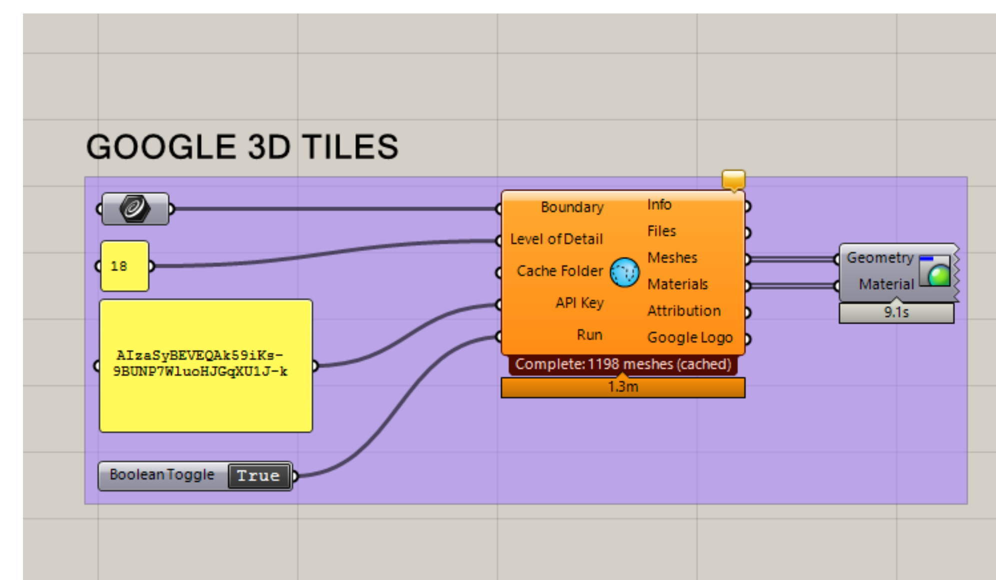
Creating the topo mesh was next, along with extracting contours from the site to prepare for creating a laser cut file. The topo mesh is **extremely accurate, which is a major strength of LiDAR** and OSM data. This allows for an accurate 3D model to design on as a base. Extracting the contours from the precise topographic mesh is easy, and **allows for a streamlined process in gathering physical site information** in a digital model.





google 3D tiles

Adding the google 3D tiles to the topographic mesh adds another layer of complexity and information, helping **designers refer directly to the site within a 3D model**. Using this photorealistic imagery can help designers see and understand their designs in a multidimensional world that directly relates to the site.



assignment 01 - Site Modeling with LiDAR and Physical Fabrication

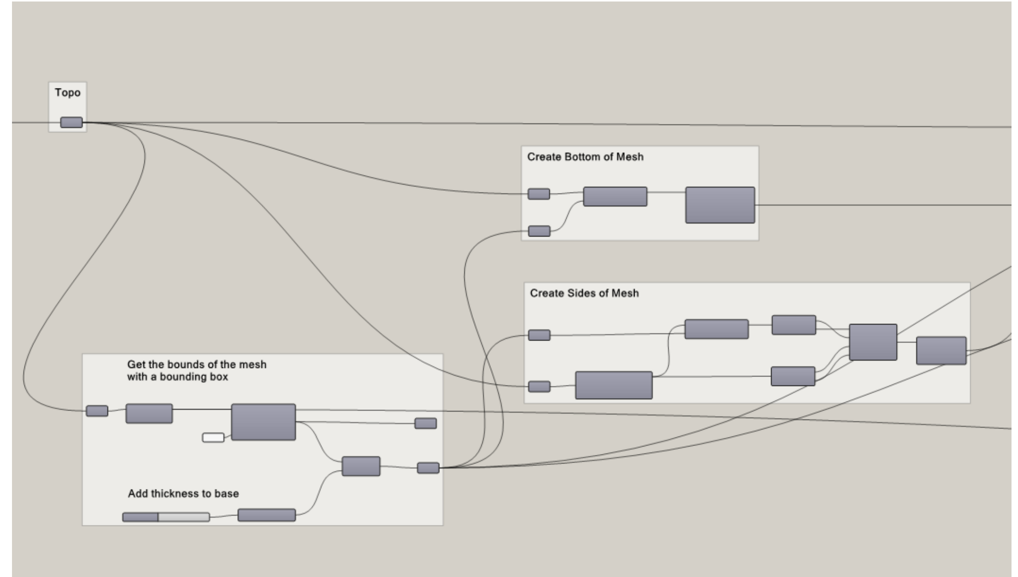
laser cut file

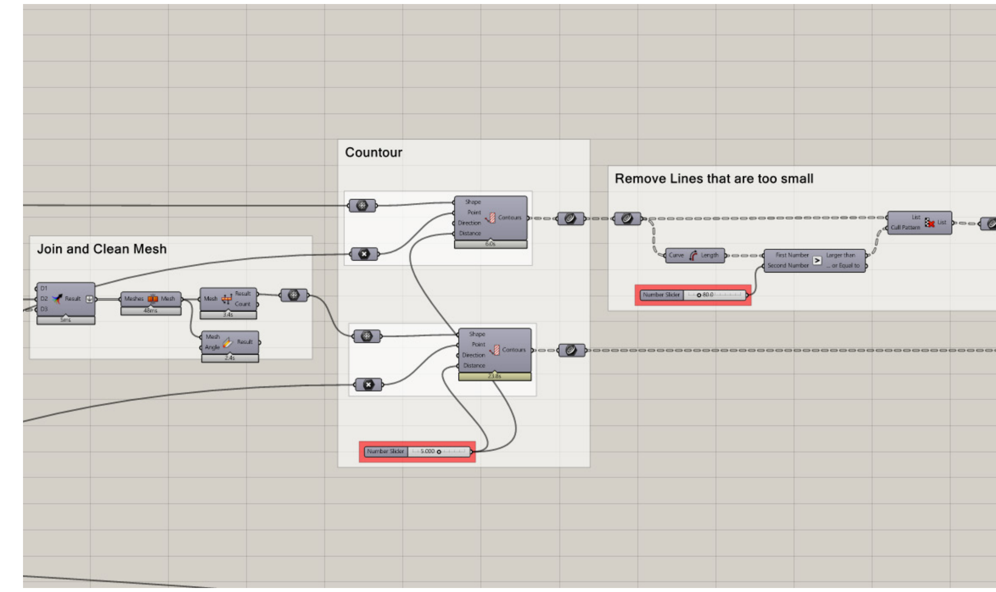
Using LiDAR data and information helps create extremely **accurate laser cut files for physical modeling**. Establishing a bounding box to hold the mesh in and extrude allows for the **extraction of contours at specific heights**, which are the layers of a laser cut file. This process not only created very accurate layers, but also **allows for a wide range of layers**, depending on how far apart the contour layers are set.

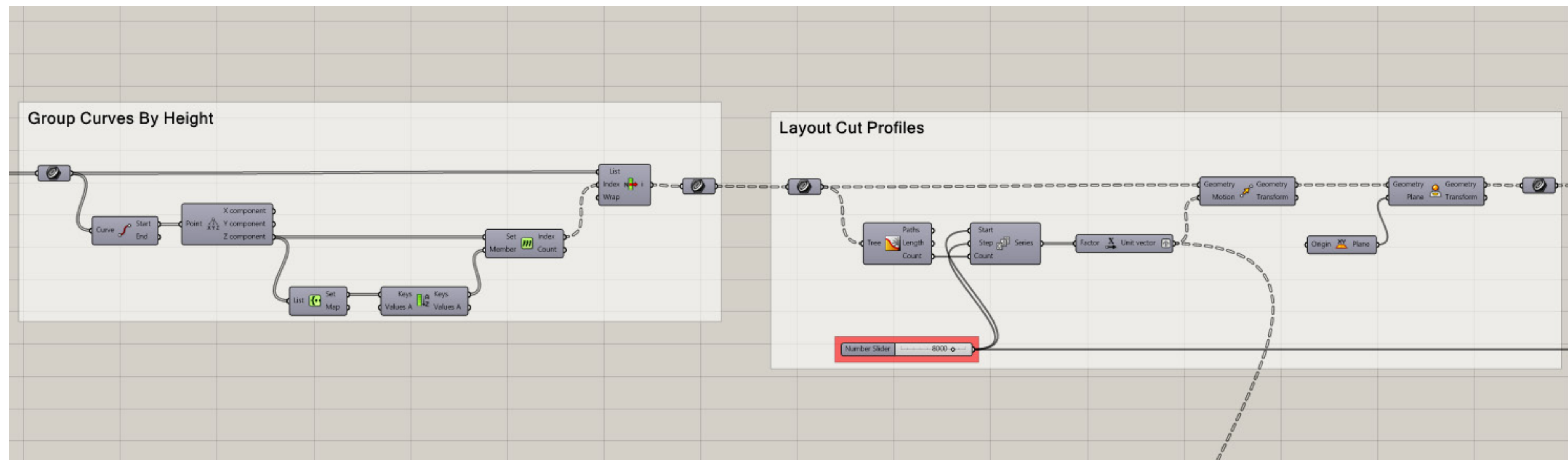
3. reflection

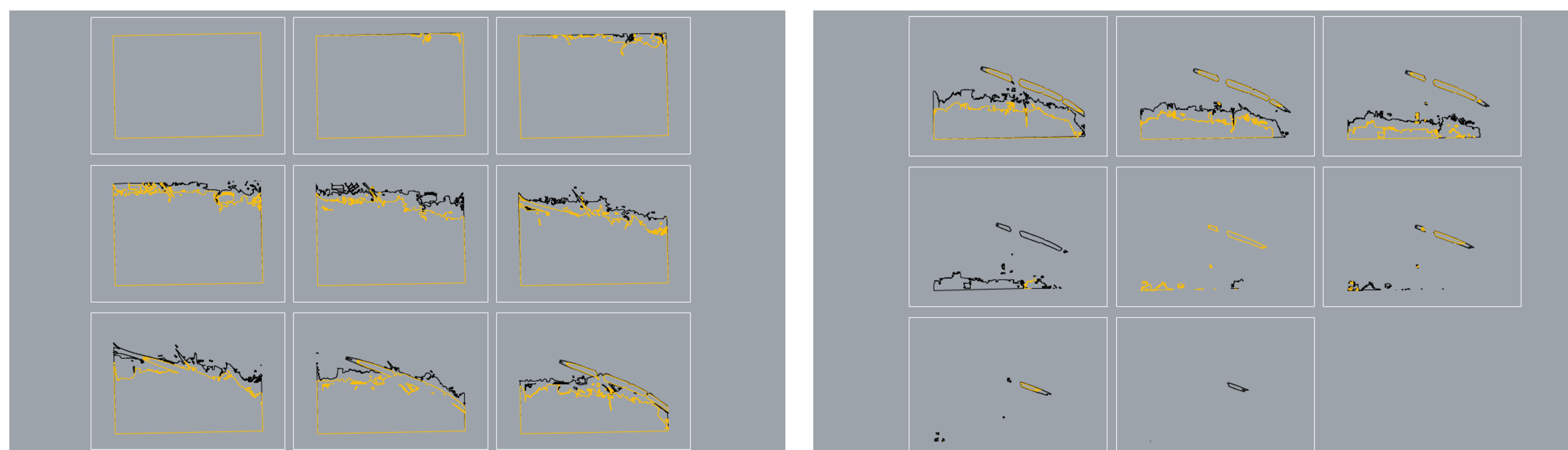
LiDAR technology provides **extremely accurate data** that can be utilized to create both physical and digital site models. Gathering LiDAR data and interpreting it through various softwares is a relatively **quick process**, and can create a **streamlined process to gather topographic data for virtually any site for any project.** The geometry that LiDAR provides has little to no human errors due to point cloud data, **ensuring the most accurate representation of topography**.

Although there are many strengths to LiDAR data, there are also a few limitations, particularly in the data acquisition and understanding. LiDAR data can be complex, and some **software needed to interpret the data can be difficult or confusing to use**. Also, a larger area needs much more data, which can take a long time to interpret and calculate when forming a topographic surface. While not impossible in the slightest, it does **require an above average level of knowledge and computational power** to utilize the information in a proper way.









assignment 01 - Site Modeling with LiDAR and Physical Fabrication